Sibley School of Mechanical and Aerospace Engineering

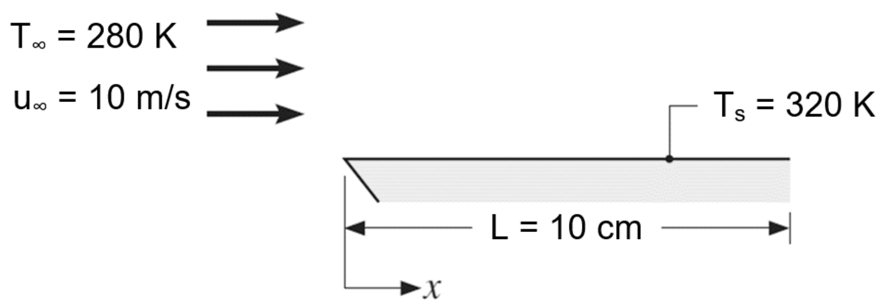
MAE 3240 Heat Transfer

Spring 2026

Problem Set #6

Posted: Monday, Mar 23 **Due: Tuesday, Apr 7, 11:59 pm**

1. (40 pt) A flat plate with a length of L = 10 cm and a surface temperature of T_s = 320 K is exposed to a coolant flow with a temperature of T_∞ = 280 K and an incoming velocity of u_∞ = 10 m/s. Properties of the coolant can be determined at the film temperature $T_f = (T_s + T_\infty)/2$. The origin of the x-coordinate is located at the leading edge of the plate, as shown in the figure below.

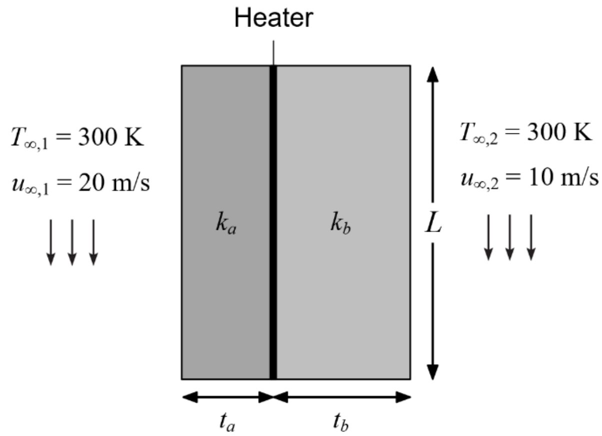


(a) If the coolant is air at the atmospheric pressure, calculate the local heat transfer coefficient h_x at $x = L/2$. In addition, calculate the average heat transfer coefficient across the plate $\overline{h_L}$. *Hint: you may find air properties from Table A.4 of the textbook.*

(b) If the coolant is water, calculate the local heat transfer coefficient h_x at $x = L/2$. In addition, calculate the average heat transfer coefficient across the plate $\overline{h_L}$. *Hint: you may find water properties from Table A.6 of the textbook.*

(c) Discuss which type of coolant provides the higher heat transfer performance at the same flow velocity and briefly explain the reason based on your calculations.

2. (30 pt) A thin film heater is inserted between two blocks *a* and *b*, as shown in the figure on the right. The heater has a square shape with side length L = 2 cm. The cross-sectional areas of two blocks are identical to the heater area. Block *a* has a thermal conductivity of k_a = 5 W/m·K and a thickness of t_a = 1 cm. Block *b* has a thermal conductivity of k_b = 16 W/m·K and a thickness of t_b = 2 cm. The left surface of block *a* is exposed to an air flow at a temperature of $T_{\infty,1}$ = 300 K and an incoming velocity of $u_{\infty,1}$ = 20 m/s. The right surface of block *b* is exposed to an air flow at a temperature of $T_{\infty,2}$ = 300 K and an incoming velocity of $u_{\infty,2}$ = 10 m/s. Please use the following air properties for the calculations: ν = 15.89×10^{-6} m^2/s, k_f = 26.3×10^{-3} W/m·K, and Pr = 0.707. The heater generates a uniform heat flux of q'' = 9 kW/m^2. The heater is thin enough with a uniform temperature T_h. There is no contact resistance between the blocks and the heater. The entire system is at a steady state.



(a) Calculate the average heat transfer coefficients h_1 and h_2 of the air flows across the block *a* and the block *b*.

(b) Calculate the heater temperature T_h. Heat transfer across the block *a* and the block *b* can be modeled as a 1D system by taking h_1 and h_2 as the inputs.

3. (30 pt) A long cylindrical wire has a diameter of D = 1 cm and an initial temperature of T_i = 350 K. At the moment t = 0 s, the wire is immersed into a coolant reservoir at a temperature of T_∞ = 300 K. A coolant with a velocity of V = 0.01 m/s flows through the wire to reduce its temperature. The coolant flow has a kinematic viscosity of ν = 15×10^{-6} m^2/s, a thermal conductivity of k_f = 0.1 W/m·K, and a Prandtl number of Pr = 1. The wire has a thermal conductivity of k = 300 W/m·K, a heat capacity of c = 300 J/kg·K, and a density of ρ = 5000 kg/m^3.

(a) Calculate the average heat transfer coefficient h of the coolant flow across the cylindrical wire.

(b) Calculate how long it will take for the wire to reach 5 K above the coolant reservoir temperature.